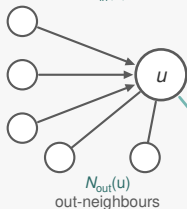


directed graph

in-neighbours
 $N_{in}(u)$



Model cells ● gate ● time

gat_dir — —

rcgat_sym ● —

gat_time — ●

rcgat_time ● ●

Directed two-view attention encoder

2 layers, LayerNorm

in-view aggregation

multi-head GAT weights
 $\alpha^{(in)}_{uv}$ over $N_{in}(u)$

$\times g_{in}$

out-view aggregation

multi-head GAT weights
 $\alpha^{(out)}_{uv}$ over $N_{out}(u)$

$\times g_{out}$

concat

in | out views
2 directed GAT layers
+ LayerNorm
 $h_u = \text{branch}$

edge-time branch

(optional)

$$\psi_{uv} = \log(1 + \Delta D)$$

W_t mean-scatter

U (zero-init)

low-rank projection

+

embedding

$$z_u = x_u + U h_u$$

cosine

$$s = \cos(z_u, z_v)$$

BCE loss

1 pos, 10 neg

region descriptor

$$\rho(u) \in R^{13}$$

8 structural + 5 temporal
on the fit graph + dates

region-conditioned gate (proposed)

$$g_d = 1 + \tanh(w_d^T \rho(u) + b_d)$$

linear layer, zero-init; $g_d = 1$ at epoch zero

node content

x_u : TF-IDF, SVD (300-d)
content residual
(epoch-zero anchor)